Controlling Foundation Models via Internal Interventions: A Survey of Inference-Time Methods

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Abstract

Large generative models exhibit powerful yet implicit behaviors that can be systematically influenced by intervening in their internal computation at inference time. Rather than updating model parameters or altering explicit prompts, inference-time interventions modify internal activations, neuron responses, attention mechanisms, routing patterns, token embeddings, or decoding dynamics during the forward pass. These methods provide a flexible, reversible, and deployment-friendly framework for controlling model behavior post-training. This survey provides a systematic review of the field, categorizing methods into representation-level interventions (activations and neurons), attention-level steering, routing-based control, and approaches targeting token embeddings and decoding stages. We analyze these interventions across three major classes of foundation models: Large Language Models (LLMs), Multimodal Large Models (MLLMs), and Diffusion Models. Finally, we discuss the relative strengths and limitations of these techniques and outline future directions for steering foundation models.

1 Introduction

Generative foundation models, including large language models (LLMs) and multimodal large models (MLLMs), now form the core infrastructure for modern AI systems. Web-scale pretraining and scaling laws have enabled strong few-shot and in-context capabilities [4], while instruction tuning and reinforcement learning from human or AI feedback (RLHF/RLAIF) have improved alignment and usability [38]. In parallel, advances in multimodal pretraining have produced vision–language models capable of grounding, reasoning, and generation across text and images, enabling applications in perception, dialog, and embodied settings.

Despite significant advances, reliably controlling model behavior remains challenging: models may hallucinate, violate safety constraints, or deviate from user intent. The dominant solution – weight-level adaptation via fine-tuning or RLHF – is effective but expensive, slow to iterate, and often brittle under distribution shift. These limitations motivate the search for lightweight, flexible control mechanisms that can be applied after deployment without retraining or modifying the full set of model parameters.

Early work on controllable generation explored explicit conditioning signals, such as control codes [27], and "plug-and-play" methods that steer hidden states via auxiliary attribute models [10]. Recently, prompting and prompt engineering have become the primary control interface for LLMs, enabling task adaptation through natural-language instructions and few-shot demonstrations [4], and have shown widespread adoption. However, prompting is fundamentally an input-space control mechanism and can fail when instructions are ignored, difficult to express, or bypassed under adversarial or distribution-shifted inputs. Parameter-efficient alternatives such as soft prompts, prefix tuning, and low-rank adapters [20] reduce training cost but still require optimization and are not designed for dynamic behavioral control.

These limitations have catalyzed a shift toward methods that intervene directly on internal model computations during inference. From a causal perspective, transformer models define directed computational graphs where information flows through embeddings, attention modules, feedforward blocks, and residual connections before influencing final token logits. Inference-time interventions operate by targeting specific intermediate variables within this graph, modifying internal nodes or edges while holding upstream inputs and parameters fixed. By bypassing linguistic mediation, these mechanisms directly alter the causal pathways governing generation, enabling fine-grained and reversible behavioral control at deployment time.

Taxonomy and failure-mode perspective. Building on this intervention-based view, we organize prior work along two axes shown in Table 1: where an intervention is applied in the generative pipeline, and which computational mechanism it modifies. We group methods into three broad classes:
(1) Representation-level interventions, which directly edit internal activations (e.g., residual streams, neurons, or learned latent features).
(2) Attention-level interventions, which modify the model's routing over tokens/regions (e.g., reweighting attention maps, suppressing sinks, or editing head-level interactions).
(3) Decoding-level interventions, which alter the generation

procedure by reshaping token selection and sequence scoring at inference time - for example, via contrastive scoring, verifier-/critic-guided re-ranking, constrained sampling, or reward-guided decoding - while leaving the base model parameters unchanged.

Importantly, these mappings are not one-to-one: many methods blur boundaries, and control objectives can be addressed via multiple approaches. Representation-level interventions are often applied when undesired behaviors appear to be encoded in internal features (e.g., stylistic tendencies, bias-related directions, or spurious activations). Attention-level interventions are commonly used when failures arise from evidence selection or grounding(e.g., the model under-attends to relevant context or over-attends to distractors/sinks). Decoding-level interventions act as a last-mile interface: they reshape logits or re-rank candidates to enforce output-side constraints (e.g., format, repetition, verbosity) and to fuse in external scoring signals (e.g., verifiers, reward models, retrieval evidence, multimodal consistency) without modifying internal representations or model weights.

Contributions and Organization. This survey makes the following contributions:

- A taxonomy of inference-time interventions, categorized by intervention family and internal mechanism.
- A comprehensive analysis of existing methods, evaluating performance gains and the generalizability of induced behaviors across models (see Table 2).
- A discussion of future directions and open challenges for the field of internal model steering.

2 Representation Level Intervention

2.1 Activation Steering

Activation (representation-level) steering refers to inference-time methods that control model behavior by directly intervening in intermediate activations while keeping model parameters fixed. Although first explored in LLMs, the same idea extends naturally to vision-language models (VLMs): modern VLMs fuse projected image tokens and text tokens inside a shared transformer backbone, so the same intervention machinery can operate on unimodal or multimodal hidden states, including cross-modal representations.

A minimal formalization. Let f_θ be a pretrained transformer (uni or multi-modal) with parameters θ, and let $h_{l,t} \in \mathbb{R}^d$ denote the hidden activation at layer l and position t (text token or visual token). Activation steering applies an inference-time perturbation as follows:

$$h'_{l,t} = h_{l,t} + \Delta_{l,t}, \tag{1}$$

where $\Delta_{l,t}$ may be (i) a fixed additive shift (a "steering vector"), (ii) a multiplicative rescaling or mask, (iii) an input-conditioned update predicted from context, or (iv) a localized patch/transport map. This encapsulates numerous rules of thumb in a single equation while bearing down on the important constraint: θ does not change.

Residual Stream / Layer Block The most common steering approach is the residual stream (or layer/block outputs), where additive shifts are broadly effective and easy to implement. Turner et al. [50] introduce Activation Addition (ActAdd), which computes a steering direction from contrasting prompts (e.g., positive vs. negative sentiment) and applies it during decoding to control sentiment, topic, and toxicity across multiple model families. By subtracting two runs that are intentionally close in content but differ in the attribute of interest, the resulting difference aims to cancel "what the prompt is about" and retain "how the model is behaving" along the targeted dimension. Dataset-level variants stabilize this idea by averaging over many exemplars: Contrastive Activation Addition (CAA) computes mean activation differences over positive/negative datasets to steer Llama models [39], while mean-centering variants improve robustness by subtracting behavior-agnostic baselines to reduce spurious shifts [23]. A closely related perspective treats in-context learning itself as an implicit latent-space shift and makes that shift explicit: In-context Vectors (ICV) extract a single task vector from demonstration examples (e.g., via principal directions of demo-pair activation differences) and then add it across layers/tokens at inference, effectively "compressing" demonstrations into an activation-level steer that is continuously controllable in strength and composable via simple vector arithmetic [33]. Residual-stream steering has also been used to modulate reasoning traces: Azizi et al. [1] construct directions from paired verbose reasoning rationales and inject them during decoding to compress chain-of-thought (CoT) while constraining distributional drift. Beyond additive translations, multiplicative interventions rescale existing features rather than injecting new ones; Liu et al. [32] show scaling-based steering can provide comparable control with reduced interference, especially when targeting feature-selective subspaces. This directly connects to mechanistic interpretability: specialized heads/neurons can be identified and selectively modulated for greater causal control.

VLM-specific. In VLMs, interventions can occur before fusion (vision encoder features / post-projector vision tokens) or after fusion (shared multimodal residual stream). VTI [34] reduces hallucinations by injecting stabilizing latent shifts at inference time, approximating the effect of averaging vision features under mild perturbations. While SteerVLM [45] trains a lightweight add-on module conditioned on contrastive target/converse prompts to produce token-wise activation updates injected across decoder layers while keeping the base VLM frozen. Text-derived steering directions can also transfer into multimodal models: [13] extract directions from the text-only backbone and inject them into multimodal hidden states to improve visual understanding (e.g., spatial relations and counting), suggesting that some "visual" concepts are linearly addressable in shared representation spaces.

Activation-level interventions in diffusion denoisers. Diffusion models expose rich intermediate representations (U-Net/DiT feature maps and attention maps) that can be intervened on at inference time, enabling training-free control similar to activation steering in transformers. Diffusion steering can likewise be grouped by intervention locus: (i) block

Intervention Approach	Taxonomy	Related Work
Representation Level Intervention	Activation Steering	ActAdd [50], CAA [39], Mean-center activation steering [23], ICV [33], CoT compression steering [1], Activation Scaling [32], VTI [34], SteerVLM [45], Textual Steering Vectors [13], CASteer [12], ActErase [48], CountSteer [3], Activation Transport (AcT) [43]
	SAE Steering	LF-Steering [55], FGAA [46], CorrSteer [7], SAE-SSV [18], SAIF [19]
Attention Level Intervention	Token Reweighting	PASTA [58], InstABoost [17], CAAC [11], CARVE [14]
	Dynamic Control	SpotLight [51], COMPASS [44], LLMSTEER [15], SPARC [24], Qwen Look Again [8]
	Redistribution/ Sinks	StreamingLLM [54], First-token sink analysis [2], ACT [56], Sink analysis [16], VAR [25], EVAS [59]
	Activation inside Attention	DISCO [49], ITI [29]
	Module Localization	REAL/DEAL [57], Ships/Sahara [62], Head importance analyses [35, 52]
Decoder Intervention	Single-Model Objective	SimCTG [47], EGED [9]
	Logit Composition	Contrastive Decoding [30], ThinkLogit [60], Contrastive decoding improves reasoning [36]
	Multimodal Contrast	VCD [28], ICD [53], ConVis [41]
	Attention-Aware Decoding	OPERA [21], IMCCD [31]

Table 1: **Taxonomy of Steering Methods** for Multimodal AI, organized by Model Architecture.

feature maps, (ii) localized activation patching/masking, and (iii) attention-projection edits (e.g., query projections). CASteer) [12] precomputes concept-specific steering directions from averaged hidden activations and applies them dynamically to cross attention outputs to suppress undesired concepts only when present (or inject desired concepts when absent). While CountSteer [3] derives directions from contrasting correct vs. incorrect counting activations and injects step-/block-specific shifts with adaptive strength. Both these inject steering vectors into cross-attention. ActErase [48] performs training-free concept erasure targeting FFN layers and performing activation patching: it contrasts activations for source vs. target prompts, localizes concept-specific regions, and masks/replaces those entries during denoising. Activation Transport (AcT) [43] learns activation-space transport maps between source/target distributions and applies them as a training-free control mechanism for style induction and concept negation. It intervenes broadly at layer outputs - normalization layers and residual-layer outputs of models.

2.2 Sparse Autoencoder Steering

Sparse Autoencoders (SAEs) enable steering in latent feature spaces rather than dense activations. This process involves encoding activations into sparse features, intervening in the latent space, and decoding the result back into the model's original space as follows:

$$z_{l,t} = E(h_{l,t}), \qquad h'_{l,t} = D(z_{l,t} + \Delta z_{l,t}) \,. \tag{2}$$

Here, $h_{l,t} \in \mathbb{R}^d$ is the original hidden activation (layer l, token position t), E and D are the SAE encoder/decoder, and $z_{l,t} \in \mathbb{R}^m$ is a sparse latent code. Steering is performed by editing a small subset of latent features via $\Delta z_{l,t}$ (e.g., boosting/suppressing selected coordinates); the edited dense activation $h'_{l,t}$ is then substituted for $h_{l,t}$ during inference while keeping the base model weights fixed. When reconstruction is accurate, this is approximately equivalent to adding a structured perturbation in model space, i.e., $h'_{l,t} - h_{l,t} \approx D(\Delta z_{l,t})$. By restricting interventions to a small set of salient latent dimensions, SAE-based steering aims to reduce polysemantic interference and improve interpretability relative to dense-state steering. LF-Steering [55] steers individual SAE features and improves semantic consistency and robustness across NLU/NLG benchmarks. In addition, another approach, FGAA [46], combines contrastive analysis with SAE features to construct optimized steering vectors that achieve significant improvements over activation-based steering approaches. CorrSteer [7] selects SAE features correlated with task correctness to improve QA and reasoning benchmarks. SAE-SSV [18] restricts supervised steering to small task-relevant SAE subspaces, enabling precise intervention.

SAIF [19] uses sparse autoencoders as a lens for interpreting instruction-following and then turning that interpretation into a steering mechanism: by analyzing SAE latent activations, it identifies a distinct subset of instruction relevant latent features, shows that these features are semantically aligned with the instruction content, and demonstrates that manipulating them can causally improve instruction adherence. It highlights design choices that matter for SAE steering efficacy, including careful feature identification/selection, the importance of interventions near the final layers, and sensitivity to where the instruction appears in the prompt.

O'Brien et al. [37] steer refusal behavior by amplifying SAE features associated with refusing unsafe requests. Their results suggest SAE steering can increase robustness to both single-turn and multi-turn jailbreak attempts, but also expose a key caveat: strengthening refusal through SAE features

can incur systematic capability degradation on downstream benchmarks, even for benign inputs. This points to an important open problem for SAE-based safety interventions-safety-relevant features may be more entangled with general capabilities than expected, so naive characteristic amplification can trade off utility for safety.

Rather than directly steering by turning SAE features up or down, SAE-TS [6] uses SAEs as a measurement and targeting tool for steering vectors. The core idea is to evaluate how any candidate steering vector changes activation-space behavior in terms of SAE feature effects, and then construct a steering vector that deliberately targets specific SAE features while minimizing unintended side effects. Empirically, SAE-TS aims to produce steering vectors that better balance "strength of steering" with coherence/quality compared to contrastive baselines (e.g., CAA) and direct SAE-feature steering.

2.3 Discussion

Representation-level steering is a way to change model behavior without retraining. Instead of only shaping the next-token choice at the end, it nudges the hidden state towards the intended direction for the (Eq. 1). That matters because many problems look like the model slipping into the wrong mode: becoming generic, overly verbose, overly cautious, inconsistent in refusal, or, in VLMs, hallucinating. A small shift to intermediate activations can push the computation back toward a better trajectory and influence many downstream tokens. Often the weights stay fixed, and the intervention is reversible, and while strength is easy to tune by scaling $\Delta_{l,t}$. From Table 2 we observe that steering based apporaches show improvement in performance across various tasks.

Another advantage is that these interventions often come with an idea about what is being changed. Contrastive directions make the control signal explicit ("what differs between two matched prompts or datasets?"), and SAE-based methods turn dense activations into a sparse set of features you can inspect and edit (Eq. 2), which can reduce interference and make debugging easier. The same idea also transfers naturally across settings: in VLMs, you can intervene either before fusion (cleaning up visual evidence) or after fusion (steering the shared multimodal stream), and diffusion models admit similar training-free edits on feature maps and attention-like representations. The open problem is reliability: picking layers/positions that generalize across prompts, avoiding spurious directions, deciding when to activate the steer (rather than applying it everywhere), and preventing side effects for various tasks such as style, safety, grounding, etc.

3 Attention Level Intervention

Attention steering refers to inference-time interventions that bias a model's attention computation so generation aligns better with a user goal (e.g., follow instructions, ground on evidence, reduce unsafe content, or improve serving efficiency) modifies the model's information routing mechanism—for example by reweighting attention over highlighted tokens, adding biases to attention logits, or editing key/query/value representations—while keeping base weights fixed.

Token reweighting. These approaches directly reshape attention mass toward selected tokens or spans, typically without closed-loop feedback. In LLMs, this is often instantiated as prompt highlighting or instruction emphasis: PASTA [58] reweights attention toward user-highlighted spans, while InstABoost [17] uses a minimalistic rule that boosts attention to instruction tokens. In VLMs, the same mechanism targets visual grounding by reweighting attention over visual tokens or regions: CAAC [11] mitigates hallucination by balancing attention across visual tokens and adaptively reinforcing visual grounding based on token-level confidence, and CARVE [14] refines attention by contrasting task-specific versus general attention maps to suppress visual noise and emphasize task-relevant evidence. SEKA/AdaSEKA fits here as a systems-oriented variant: rather than editing attention weights post-hoc, it steers via attention-input edits (keys) that still yield a static "emphasis" effect.

Dynamic control (state-dependent, progressive, gated, or closed-loop). Dynamic methods make steering conditional on the decoding state-changing strength over time, triggering only when needed, or using feedback signals to regulate attention. SpotLight [51] represents time-varying prompt highlighting, while COMPASS [44] frames grounding as feedback control (steering attention toward evidence when hallucination risk rises). LLMSTEER [15] adapts steering to long-context serving, especially under context reuse, by applying an attention policy tuned to deployment constraints rather than a static prompt rule. In VLMs, analogous concerns arise during long-form reasoning and captioning where visual attention can drift or fade: SPARC [24] progressively recalibrates attention to preserve visual grounding over long generations; cross-attention gating approaches trigger realignment when the model appears to rely too little on visual evidence; and Qwen Look Again [8] explicitly re-attends via reflection-like steps and visual-token operations.

Redistribution (attention budget recycling/sinks). A common long-context pathology is attention sinks - tokens (often BOS, but sometimes other low-semantic positions) that attract disproportionate attention and waste routing budget. StreamingLLM shows that preserving the KV cache of initial sink tokens largely restores streaming performance under windowed attention, and proposes adding a dedicated sink token during pretraining to better support deployment [54]. However, sinks are not always pure waste: often, the first-token sink can play a stabilizing role by helping Transformers avoid over-mixing, so indiscriminate suppression can hurt quality [2]. This motivates selective recycling methods such as ACT, which detects sink tokens (including beyond BOS) and downscales their attention in chosen heads/layers while reallocating the freed mass to non-sinks with normalization, improving accuracy without weight updates [56]. Complementary analysis suggests that many sinks behave like key-bias artifacts, storing attention scores while contributing little to value computation, supporting the view that some sink mass is recyclable surplus [16].

The idea can be extended to multimodal KV sets by identifying fixed-location visual sink tokens (linked to large sink-dimension activations) that can often be removed with little loss, then redistributing their attention within image-centric heads to strengthen evidence use and reduce hallucinations

[25]. EVAS [59] similarly links hallucinations to sink structure across layers and increases image focus by broadcasting the attention map from the densest sink head to other heads in the same layer.

Deep inside attention (activation steering inside attention modules). Some methods intervene not on attention weights but within attention submodules-editing Q/V representations or attention-head activations. We include these in attention steering but explicitly label them as activation steering within attention modules, because the interventions are activation-level yet attention-located. In LLMs, DISCO [49] injects steering signals into Q/V spaces to obtain more disentangled control; ITI [29] shifts activations in selected attention heads to steer behavioral traits such as truthfulness; and SADI makes steering vectors semantics-adaptive (input-dependent), addressing brittleness of fixed directions.

Module localization and intervention enablers (LLMs). Because steering success depends strongly on where to intervene, a complementary line of work focuses on localizing behavior-relevant modules (heads/layers) and providing enablers for targeted control. On the LLM side, DEAL/REAL [22] proposes a principled framework to identify behavior-relevant attention modules (heads/layers) for more reliable inference-time steering, rather than relying on ad hoc heuristics [57]. For safety, Ships/Sahara identifies safety-critical attention heads via head-level importance scoring and attribution, enabling targeted monitoring, ablation, or steering of a small subset of modules [62]. More broadly, this direction connects to classic attention-head importance/pruning analyses, which show that only a subset of heads are causally important for performance or specific behaviors [35, 52]. Limitations: localization can require offline analysis or auxiliary training; identified modules may not transfer perfectly across domains/prompts; and making critical modules explicit can increase the attack surface in safety settings (e.g., by targeting identified safety heads) [62].

Extension to VLMs. In VLMs, analogous analyses identify "image-centric heads" and "sink tokens" that absorb disproportionate attention, creating actionable loci for interventions such as attention budget redistribution (e.g., VAR) [26]. Mechanistic studies trace where and how visual and linguistic information are integrated across layers, helping isolate candidate cross-modal integration/failure loci for localization or editing [61]. TAB [42] provides an architectural enabler: an explicit attention bottleneck that exposes a single, editable attention map, making user intervention more interpretable and causally actionable (by modifying the architecture).

4 Limitations of Attention Steering

Attention steering addresses context utilization failures: cases where the model already has the relevant information in its current context (prompt, retrieved evidence, or image tokens), but fails to use it because attention is misallocated. In practice this shows up as (i) **instruction underuse**, where instruction tokens lose salience as decoding progresses and the model defaults to generic continuation; (ii) **evidence neglect**, where cited passages or grounded visual regions are present but weakly attended, leading to unfaithful or hallucinatory generation; (iii) **long-context pathologies**, where attention budget is wasted on structural tokens (e.g., sinks, boilerplate prefixes) rather than task-relevant spans; and (iv) in VLMs, modality drift, where the model increasingly relies on language priors instead of image evidence during long-form generation. Attention-level interventions are attractive here because they offer a per-request, inference-time handle to reroute information flow without updating base weights, and thus complement other approaches (e.g., retrieval or prompting) by increasing the probability that available evidence is actually consumed by the decoder.

5 Decoding Intervention

Transformer models often assign non-trivial probability mass to irrelevant tokens for various reasons (e.g., repetition loops, generic high-probability phrasing, premature stopping) or due to competing signals (e.g., language priors vs. visual evidence). Decoding intervention provides an intervention approach that changes which tokens are selected during generation by modifying the token scores used by the decoder (i.e., logits/probabilities) and/or the decoding objective (e.g., re-ranking candidates in beam search), without updating the model parameters. We can describe this via an equation. Let $\mathbf{z}_t$ denote the base model logits at step t. Decoding intervention forms adjusted logits $\tilde{\mathbf{z}}_t$ before sampling/argmax as follows:

$$\tilde{\mathbf{z}}_t \;=\; \mathbf{z}_t \;+\; \sum_k \lambda_k \, s_k(x, y_{<t}), \tag{3}$$

where each $s_k(x, y_{<t})$ is an additive steering score produced by mechanism k (e.g., a contrastive score, a penalty term, or evidence-based adjustment), and λ_k is a scalar weight that controls how strongly that mechanism influences decoding. Setting $\lambda_k = 0$ disables signal k, while larger $|\lambda_k|$ yields stronger steering.

Single-model objective reshaping. The most lightweight family reshapes the decoding objective using only signals available within a single forward pass (or within a single model), targeting generic decoding failures such as repetition and blandness. Contrastive Search [47] (often associated with SimCTG) augments token selection with a representation-similarity penalty, discouraging candidates that are too similar to previously generated states, and thereby improving coherence and diversity without auxiliary training. Moving from general quality to factuality-oriented signals, Entropy Guided Extrapolative Decoding [9] leverages internal layer-wise distributions and entropy-based criteria to calibrate generation; the method remains training-free but typically assumes white-box access to intermediate-layer outputs.

Logit composition across pretrained models. Whereas single-model methods reshape decoding using only the base model's internals, a second family composes distributions from multiple pretrained models, using logit arithmetic to amplify desirable behaviors and suppress undesired ones.

Contrastive Decoding [30] contrasts an "expert" model against a smaller "amateur" model (expert minus amateur) under a plausibility constraint, often yielding higher-quality generations at the cost of an additional forward pass.

Task	Method	Model	Benchmark	Metric	Base	Steered
Representation-Level Steering (LLMs)						
Toxicity reduction	ActAdd [50]	GPT-J 6B	RealToxicityPrompts	Mean toxicity	0.075	0.07
Sentiment control	ActAdd [50]	GPT-J 6B	IMDb	"Success" rate	0.63	0.79
Capability audit	CAA [39]	Llama-2-7B	MMLU (subset/proxy)	Avg. prob(correct)	0.51	0.5
Function vectors	Mean-center [23]	GPT-J-6B	6 FV tasks (paper)	Accuracy (%)	29.2	45.7
Detoxification	ICV [33]	Llama-7B	ParaDetox	Toxicity (%)	71.6	39.54
Style transfer	ICV [33]	Llama-7B	GYAFC (formality)	Formality (%)	17.54	48.3
Semantic consistency	LF-Steering [55]	Llama2-7B-Chat	RobustBOOLQ	Accuracy (%)	46.4	66.4
Multi-goal steering	FGAA [46]	Gemma-2-2B	9 steering goals (paper)	Mean BCS	0.2201	0.4702
Sentiment steering	SAE-SSV [18]	Llama3.1-8B	Steering eval (paper)	SR (%)	45.6	63.2
Representation-Level Steering (VLMs/LMMs)						
Hallucination mit.	VTI [34]	LLaVA-1.5	CHAIR(↓)/ POPE/ MMHal	CHAIRs/Acc/Avg. score	51 / 79.8 / 1.99	35.8 / 86.5 / 2.9
Safety steering	L2S [40]	LLaVA-v1.5	MMSafetyBench	Unsafe-score	0.234	0.057
Hallucination mit.	L2S [40]	LLaVA-v1.5	POPE (Random)	Accuracy (%)	82.73	86.46
Hallucination mit.	L2S [40]	LLaVA-v1.5	COCO (500 imgs)	CHAIRs (%)	17.31	16.1
Hallucination mit.	SteerVLM [45]	LLaVA1.5-7B	OHD (POPE overall)	Accuracy (%)	84.5	86.4
Topic control	SteerVLM [45]	LLaVA1.5-7B	5 steering vectors (paper)	Judge score (0–1)	0.55	0.71
Representation-Level Steering (Diffusion)						
Concept erasure	CASteer [12]	Stable Diffusion	I2P / COCO-30K	CLIP-30K / FID-30K (↓)	31.34 / 14.04	30.09 / 13.02
Attention-Level Steering (LLMs)						
Multiple tasks	PASTA [58]	LLaMA-7B	Multiple Bench.	All (avg)	73.45	95.46
Multiple tasks	PASTA [58]	GPT-J	Multiple Bench.	All (avg)	59.65	85.22
Instruction following	SpotLight [51]	Qwen2.5 7B	IFEval	Prompt / Instr. acc	0.47 / 0.59	0.54 / 0.66
Many-instruction following	SpotLight [51]	Qwen2.5 7B	ManyIFEval	Prompt / Instr. acc	0.20 / 0.63	0.26 / 0.70
Attention-Level Steering (VLMs/LMMs)						
Hallucination mit.	CG-VLM [5]	LLaVA	POPE (COCO)	F1	79.5	86.1
Hallucination mit.	CG-VLM [5]	LLaVA	CHAIR	CHAIRi (↓)	9.3	6.5
Hallucination mit.	VAR [25]	LLaVA-1.5-7B	POPE (all)/MMHalBench	F1	84.76	85.87
Hallucination mit.	Qwen [8]	Qwen2.5-VL-Instruct	CHAIR / POPE / MMHal	mixed	9.4 / 37.1 / 88.7 / 3.68	3.7 / 9.8 / 90.2 / 3.82
Decoding Intervention						
Reasoning accuracy	CD for reasoning [36]	LLaMA-65B	GSM8K / HellaSwag	Accuracy (↑)	51.0 / 84.2	56.8 / 88.0
Long-CoT elicitation	ThinkLogit [60]	Qwen2.5-32B (target) + 1.5B guider	AIME25 / AMC23 / MATH L5 / Avg	Avg@8 accuracy (↑)	8.3 / 57.2 / 44.7 / **32.3**	19.2 / 62.2 / 55.3 / **40.2**

Table 2: **(Comparative Taxonomy and Comparison.)** Performance summary of activation,attention and decoding intervention based on the following factors: task, method, language model, dataset, and availability of metrics across.

ThinkLogit [60] follows a similar principle: it uses a separate guider distribution to steer a larger model toward longer-form reasoning behaviors purely at inference time. Empirically, this general mechanism can also affect reasoning accuracy: O'Brien and Lewis [36] report that contrastive decoding can improve performance on reasoning tasks relative to standard decoding, reinforcing that decoding-time choices may change not just style but also error modes.

Multimodal contrast via controlled perturbations and consistency signals. In VLMs/MLLMs, hallucinations often arise from an imbalance between visual grounding and strong language priors. A natural training-free remedy is to form two output distributions under controlled perturbations and then contrast them to suppress tokens that persist under the perturbation (interpreted as prior-driven). Visual Contrastive Decoding (VCD) [28] contrasts logits obtained from the original image and a distorted image to reduce object hallucinations. Instruction Contrastive Decoding (ICD) [53] performs an analogous contrast in the instruction space by pairing the original prompt with a "disturbance" instruction designed to elicit hallucinations, and subtracting the disturbance-induced distribution. ConVis [41] strengthens the contrastive signal by using a text-to-image model to reconstruct images from candidate captions and contrasting distributions between the original and reconstructed images, trading off improved semantic consistency against substantially higher inference cost.

Attention-aware decoding penalties and cross-modal calibration. Often, some multimodal methods derive steering signals from attention diagnostics or cross-modal correlation measurements while still applying the intervention at the decoding stage. OPERA [21] introduces an over-trust penalty and a retrospection-allocation mechanism (implemented within beam-search-style decoding) to mitigate hallucinations without retraining. IMCCD [31] calibrates inter-modality correlation by constructing a counterfactual ("distorted") distribution through masking value vectors associated with cross-modal attention patterns, and then applying contrastive decoding to downweight hallucination-prone tokens. These methods represent a broader hybrid trend: attention-derived signals can be used to define penalties or counterfactual distributions, but the final control lever remains the token-selection rule.

Discussion. Decoding intervention is useful because it fixes a very common issue: the model often can produce a good next token, but standard decoding still picks a worse one. This happens because decoding tends to follow short-term patterns-like repeating phrases, falling back to safe/generic wording, or ending too early or because it leans too much on certain signals from language. In VLMs, it does not use signals from the image.

Equation (3) captures the basic idea: before choosing the

next token, we adjust the token scores $\tilde{\mathbf{z}}_t$ (or re-rank candidates). That lets us add simple extra preferences at the exact point where the decision is made—penalize repetition, reward diversity, favor outputs consistent with evidence, or reduce multimodal hallucinations via contrastive signals. This is easy to plug into existing sampling or beam search, easy to turn on/off, and can be applied without changing model weights during inference.

6 Future Directions and Open Problems

Widespread adoption of representation steering-based approaches still remains limited, largely due to **reliability**: effects can vary sharply across inputs, include "anti-steerable" cases, and may degrade truthfulness/safety because hidden states are polysemantic and causally entangled. While reliability remains an open direction, another area of improvement can be on how we build the contrasts used to find steering directions. Instead of relying on weak or noisy "before vs. after" examples, we should design better baselines and more systematic contrast sets, so the steering direction captures the intended effect rather than accidental correlations. When and where to apply control remains challenging and inconsistent across the literature: different papers recommend different token positions and layers (for LLMs), or different timesteps and spatial regions (for diffusion). Because the "right" intervention point is highly context-dependent, applying control too broadly can cause unwanted side effects like artifacts, erasure, or degraded truthfulness/safety, while applying it too narrowly may fail to steer at all.

Attention-level interventions. Attention interventions primarily change where the model routes information, not what it knows. This means they work best when errors come from misrouting, but they have limited impact when the bottleneck is missing knowledge or weak reasoning. A future direction is diagnosis and gating: build lightweight tests that estimate whether a failure is routing-driven (so attention control is likely to help), and trigger interventions only in those cases. Given that static emphasis can be brittle and over-focus on the wrong span/region, the next step is to move towards strengthening attention when the highlighted evidence agrees with the prompt and the model is confident, and relax it when evidence conflicts or uncertainty is high. Long-context results add a further constraint: "attention sinks" are not always waste and can sometimes stabilize computation, so the goal is not blanket suppression but selective recycling, i.e., downscaling only surplus sink mass in specific heads/layers and redistributing it while preserving normalized attention structure. In multimodal models, same principle suggests architecture-aware policies that identify and prune/redistribute fixed-location visual sinks while prioritizing image-centric heads to improve grounding. Deeper edits inside attention (Q/K/V or head outputs) can offer more precise control but are riskier; making them practical requires constrained interventions (bounded strength, automatic tuning, and side-effect monitoring for fluency/style drift) and better module localization to decide where to intervene. Localization is especially important in safety settings, where exposing "critical heads" can increase the attack surface, motivating secure monitoring/localization practices in deployment.

Decoding-level interventions Decoding intervention provides a model-agnostic control layer: rather than changing weights, we resample token scores at generation time and can compose multiple objectives through logit arithmetic. One of the bottlenecks in decoding interventions is efficiency. It often requires extra computation (multi-pass contrast, reranking, counterfactual distributions, or reconstruction-based checks), so a future direction is budgeted, risk-aware decoding - use cheap, single-pass signals by default and trigger expensive checks only when hallucination/constraint-violation risk rises, supported by caching and approximate scoring to minimize extra forward passes. Another challenge is stability; most decoding interventions operate token-by-token, providing only soft control. If the steering is too aggressive - for example, by heavily penalizing "unlikely" tokens - it can accidentally suppress rare-but-correct words, hurt fluency, or distort calibration. A more robust approach is to make the mixing weights λ_k state-dependent: apply stronger steering when the model is uncertain or the risk of hallucination/constraint violation is high, and back off when the evidence is clear. In addition, lightweight sequence-level safeguards-such as a small verifier/critic or constrained search-can help enforce global properties like faithfulness, safety, and structure over longer horizons rather than only locally. Finally, decoding control is fundamentally bounded by what the base model proposes. If the correct continuation has negligible probability under the model, logit shaping has limited leverage. This motivates hybrid pipelines that (i) improve the proposal distribution upstream (e.g., via representation or attention interventions), and (ii) use cheaper but more causally meaningful counterfactual signals in multimodal settings (e.g., perturbation- or consistency-based contrasts), while avoiding "double counting" correlated cues from attention-derived diagnostics.

7 Conclusion

In this survey, we organize the broad literature into a coarse-grained taxonomy across three primary intervention levels. Because work on inference-time control is distributed across disparate research communities—often employing inconsistent terminology and distinct intervention points—systematic comparison remains challenging. To address this, we first identify and clarify the underlying causal mechanisms within the transformer architecture, categorizing methods into representation-level (activations and neurons), attention-level (steering and redistribution), and decoding-level (contrastive scoring) interventions. By unifying these approaches, we provide a consistent framework for analyzing how localized modifications influence global model behavior across LLMs, MLLMs, and diffusion models, offering a roadmap for future research on steering foundation models.

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